

Contest - III (Full Syllabus)

A. Balanced String

1 s., 256 MB

A string is called *balanced* if the number of characters *a* is equal to the number of characters *b*.

You are given a string consisting only of characters *a* and *b*. Determine whether it is balanced.

Input

The first line contains a string *s* ($1 \leq |s| \leq 2 \cdot 10^5$), consisting only of characters *a* and *b*.

Output

Print YES if the string is balanced, otherwise print NO.

input
abab
output
YES

input
aaaaa
output
NO

B. Thala For A Reason

1 s., 256 MB

In a stadium in 100xCity, some people decided to block certain gates using barricades.

They followed a simple rule:

- If the sum of digits of a gate number is exactly 7, the gate remains open.
- Otherwise, the gate is blocked.

Given *T* gate numbers, determine the status of each gate.

Input

The first line contains an integer *T* ($1 \leq T \leq 100$) — the number of gate numbers to check.

The next *T* lines each contain an integer *R* ($1 \leq R \leq 10^9$) — the gate number.

Output

For each gate number, print on a separate line:

Thala for a reason — if the gate is open (sum of digits = 7)
Blocked for no reason — if the gate is blocked (sum of digits $\neq 7$)

input
5 7 16 34 70 1234
output
Thala for a reason Thala for a reason Thala for a reason Thala for a reason Blocked for no reason Blocked for no reason

C. Laxmi Chit Fund

1 s., 256 MB

Once upon a time in 100xCity, during New Year, several residents decided to invest money in Laxmi Chit Fund, hoping to double it in 21 days. You are the owner of Laxmi Chit Fund and want to analyze the investments made during the festive season.

Given the investment amounts, determine:

- Total money invested
- Maximum investment
- Minimum investment
- Average investment

Note: The average investment must be the floor value of the exact average. For example, if the average is 7.8, you must print 7.

Input

The first line contains an integer *N* ($1 \leq N \leq 10^5$) — the number of investors.

The second line contains *N* integers *m*₁, *m*₂, ..., *m*_{*N*} ($1 \leq m_i \leq 10^9$) — the amount of money each investor invested.

Output

Print four integers separated by spaces:

Total Max Min Avg

Where:

- **Total** — total money invested
- **Max** — maximum investment
- **Min** — minimum investment
- **Avg** — average investment (floor value)

input
5 1000 2000 1500 3000 2500
output
10000 3000 1000 2000

D. Mogambo's Happy Hour

1 second, 256 megabytes

Mogambo is going through a tough time after his breakup. Now he only becomes happy when the clock shows a **good time**.

A **good time** is defined as a time where the hour and minute are the same (i.e. *HH* = *MM*).

Whenever the clock shows a good time, Mogambo laughs like crazy and becomes happy. Otherwise, he is sad and depressed.

Given the current time in *HH* : *MM* format (24-hour clock), check if Mogambo is happy or sad.

Input

A single line containing the time in *HH* : *MM* format.

- $00 \leq HH \leq 23$
- $00 \leq MM \leq 59$

Output

If the time is good (*HH* = *MM*), print: Mogambo is happy

Otherwise, print: Mogambo is sad

input
11:11
output
Mogambo is happy

input
23:32
output
Mogambo is sad

E. Toggle Me!

1 second, 256 megabytes

You are given a string s consisting of uppercase and lowercase English letters.

Your task is to transform the string so that:

- Every lowercase letter becomes uppercase.
- Every uppercase letter becomes lowercase.

Input

The first line contains a string s ($1 \leq |s| \leq 10^5$) consisting only of English letters ($a-z, A-Z$).

Output

Print the transformed string.

input
HeLLo
output
hEllO

input
aBc
output
AbC

F. Sum Along and Beyond Diagonals

1 second, 256 megabytes

You are given a square 2D array of size $n \times n$.

Your task is to calculate:

- The sum of all diagonal elements (both main diagonal and secondary diagonal).
- The sum of all non-diagonal elements.

Input

The first line contains a single integer n ($1 \leq n \leq 1000$) — the size of the square matrix. The next n lines contain n integers each ($-10^9 \leq a_{i,j} \leq 10^9$).

Output

Print two integers: the sum of diagonal elements and the sum of non-diagonal elements.

input
3 1 2 3 4 5 6 7 8 9
output
25 20

input
5 7 -3 14 2 9 0 5 -6 8 11 13 -4 1 10 -7 6 12 -2 -5 4 15 3 19 -8 20
output
72 52

For the 1st Example:

Diagonal Sum: $1 + 5 + 9 + 3 + 7 = 25$

Non-Diagonal Sum: $2 + 6 + 8 + 4 = 20$

G. Crazy Rectangle

1 s., 256 MB

You are given two integers n and m . Your task is to print an $n \times m$ pattern, filling it alternately with numbers and lowercase English characters.

The rules are as follows:

- The first cell always starts with a number 1.
- Numbers increase sequentially: 1, 2, 3, ..., 9, 1, 2, ... (after 9 it wraps back to 1).
- Letters increase sequentially: $a, b, c, \dots, z, a, b, \dots$ (after z it wraps back to a).
- Filling alternates between a number and a letter.

Input

The only line contains two integers n and m ($1 \leq n, m \leq 200$).

Output

Print the $n \times m$ pattern following the described rules.

input
4 7
output
1a2b3c4 d5e6f7g 8h9i1j2 k3l4m5n

input
8 9
output
1a2b3c4d5 e6f7g8h9i 1j2k3l4m5 n6o7p8q9r 1s2t3u4v5 w6x7y8z9a 1b2c3d4e5 f6g7h8i9j

H. Interleaved Strings

1 second, 256 megabytes

You are given two strings S and T .

Form a new string by alternating characters of S and T , inserting a dash (-) between each pair. If one string is longer than the other, append the remaining characters at the end.

Input

The first line contains the string S ($1 \leq |S| \leq 10^5$). The second line contains the string T ($1 \leq |T| \leq 10^5$).

Both strings consist of lowercase English letters.

Output

Print the resulting merged string.

input
abcpqr xyz
output
a-x-b-y-c-z-p-q-r

input
abc defghi
output
a-d-b-e-c-f-g-h-i

I. Strong Password - II

1 s., 256 MB

You are given a string *S* representing a password.

Determine the strength of the password based on the following rules:

- **Strong**: contains all four of the following:
 - at least one lowercase letter (a–z)
 - at least one uppercase letter (A–Z)
 - at least one digit (0–9)
 - at least one special character (any character not in the above three categories)
- **Moderate**: contains at least three of the four categories above.
- **Weak**: contains fewer than three categories.

Input

A single line containing the string *S* ($1 \leq |S| \leq 10^5$).

Output

Print one word: *Strong*, *Moderate*, or *Weak*, representing the password strength.

input
Abc123!
output
Strong

input
abcdef
output
Weak

J. Gangster Ghafoor

1 s., 256 MB

Once upon a time in 100xCity, Gangster Ghafoor also invested in Laxmi Chit Fund. It has been more than 21 days, and he still hasn't received his money.

Now, Ghafoor is searching for the owner of Laxmi Chit Fund in 100xCity. He received an alert that the owner is somewhere on the boundary of the city.

The city is represented as an $N \times M$ grid, where each cell indicates the number of people residing in that area.

Ghafoor starts at the bottom-left corner of the grid and moves along the boundary in clockwise direction. He stops immediately when he finds the owner (a cell with value -1).

Your task is to print the values of the cells he visits along the boundary **before reaching the owner**.

Input

The first line contains two integers *N* and *M* ($2 \leq N, M \leq 50$) — the number of rows and columns.

The next *N* lines contain *M* integers each, *grid[i][j]* ($-1 \leq grid[i][j] \leq 100$) — the population of each city block.

It is guaranteed that **exactly one cell contains -1** , and it lies on the boundary of the grid.

Output

Print the values of the cells Ghafoor visits along the boundary before reaching -1 , separated by spaces.

input
3 4 5 3 2 6 1 7 8 4 9 0 -1 2
output
9 1 5 3 2 6 4 2

input
4 3 1 2 3 4 5 6 7 8 9 10 -1 12
output
10 7 4 1 2 3 6 9 12

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